

Breeze DYCOMS-II RF01: resolution and numerics

Complete: 15 cases analyzed

Data checked: 2026-09-19T11:58:17.096Z

All 15 requested four-hour cases are analyzed. This matrix does not establish faithful low-resolution DYCOMS turbulence: the four coarse WENO cases retain the wrong third-moment sign in the 600–760 m layer, and refining WENO9/no-SGS from 35 m to 10 m horizontal spacing substantially improves the third moment despite little improvement in variance-marker error. Fine WENO9/no-SGS has the smallest third-moment marker RMSE among the tested WENO cases. All three unbounded Centered2 cases develop negative total water and cannot establish physical fidelity. Cloud persistence alone is therefore insufficient. These are single-seed comparisons; the fine grid is not a proven converged solution, and the larger coarse domain confounds the resolution comparison.

Completed Centered2 runs contain physically inadmissible negative total water (sampled minima: coarse: -4.338 g/kg; canonical: -7.697 g/kg; fine: -7.401 g/kg). Their cloud and turbulence statistics are retained as diagnostics of this unbounded numerical configuration, not evidence of a physically faithful solution. The production extrema figure shows the full sampled time history; verification here means complete finite output, not physical validity.

Centered2 smoke audit: all three grids stayed finite over the first 600 simulated seconds, but canonical and fine cases produced negative total water. Across the 11 one-minute samples, minimum q_t was coarse: 1.0872 g/kg; canonical: -0.0605 g/kg; fine: -1.3679 g/kg. Passing the finite-value gate establishes neither moisture positivity nor physical validity. Production retains unbounded centered moisture; full-run extrema are reported separately in the production audit. These are smoke-test results, not completed four-hour Centered2 statistics.

At 10 m horizontal / 5 m vertical spacing with WENO9, adding Smagorinsky changes fourth-hour minimum w'^3 over 600–760 m from -0.0622 to -0.0304 m^3/s^3 , third-moment marker RMSE from 0.0159 to 0.0313 m^3/s^3 , and variance-marker RMSE from 0.0631 to 0.0769 m^2/s^2 . LWP changes from 55.48 to 53.23 g/m². Solver wall increases by 10.1%. This closure weakens the negative third moment and increases both marker errors in this paired single-realization comparison; it is not evidence against every possible closure or coefficient choice.

Breeze DYCOMS-II RF01: resolution and numerics

At 10 m horizontal / 5 m vertical resolution without SGS, WENO5 uses 70.5 solver wall minutes versus 92.8 for WENO9 (24.0% less). However, its fourth-hour third-moment RMSE against digitized in-situ markers is 0.0310 rather than 0.0159 m^3/s^3 . Its minimum w'^3 in 600–760 m is -0.0372 versus -0.0622 m^3/s^3 . Lower advection order saves wall time but weakens the negative third moment in this realization. Timings include output within the solve and exclude startup/compilation before the solve; they are not isolated kernel benchmarks.

Fine Centered2 with Smagorinsky completes in 70.2 solver wall minutes, 31.3% less than fine WENO9 with Smagorinsky. Its fourth-hour LWP is 44.17 versus 53.23 g/m^2 , and third-moment marker RMSE is 0.0382 versus 0.0313 m^3/s^3 . Negative total water prevents interpreting this cheaper run as an acceptable physical solution. Timing includes diagnostics and differing timestep counts; no GPU profiling was performed to identify the cost mechanism.

At fixed 5 m vertical spacing and fixed horizontal domain, changing WENO9/no-SGS from 35 m to 10 m horizontal spacing changes the fourth-hour minimum w'^3 over 600–760 m from -0.0126 to -0.0622 m^3/s^3 . The fine case is negative throughout 600–760 m, agreeing with the sign of the retained in-situ observations there.

For this same 35 m to 10 m comparison, unweighted RMSE against the digitized in-situ third-moment markers changes from 0.0454 to 0.0159 m^3/s^3 ; variance-marker RMSE changes from 0.0606 to 0.0631 m^2/s^2 . The third moment improves substantially, while variance does not improve by this metric. These are single realizations without observational error-bar weighting, not a convergence demonstration.

All 4 completed coarse WENO cases have positive fourth-hour w'^3 throughout 600–760 m, while the retained in-situ observations there are negative. Cloud persistence therefore does not establish faithful coarse-grid turbulence.

Coarse, WENO9 without closure: fourth-hour LWP 49.87 g/m^2 , cloud fraction 99.34%, peak w^2 0.395 m^2/s^2 , and Δq : 0.019 g/kg .

At coarse resolution without closure, WENO9 minus WENO5 changes fourth-hour LWP by 3.42 g/m^2 and peak w^2 by 0.099 m^2/s^2 . These are paired single-realization differences, not statistically established scheme effects.

At coarse resolution with WENO9, adding Smagorinsky changes fourth-hour LWP by -6.54 g/m^2 and peak w^2 by -0.038 m^2/s^2 . Third-moment marker RMSE changes from 0.0704 to 0.0595 m^3/s^3 .

Breeze DYCOMS-II RF01: resolution and numerics

Canonical, WENO9 without closure: fourth-hour LWP 61.69 g/m², cloud fraction 99.91%, peak w^2 0.521 m²/s², and Δq_t 0.003 g/kg.

At canonical resolution without closure, WENO9 minus WENO5 changes fourth-hour LWP by 2.09 g/m² and peak w^2 by 0.054 m²/s². These are paired single-realization differences, not statistically established scheme effects.

At canonical resolution with WENO9, adding Smagorinsky changes fourth-hour LWP by -9.50 g/m² and peak w^2 by -0.067 m²/s². Third-moment marker RMSE changes from 0.0454 to 0.0444 m³/s³.

Fine, WENO9 without closure: fourth-hour LWP 55.48 g/m², cloud fraction 99.95%, peak w^2 0.472 m²/s², and Δq_t 0.031 g/kg.

At fine resolution without closure, WENO9 minus WENO5 changes fourth-hour LWP by -2.06 g/m² and peak w^2 by -0.011 m²/s². These are paired single-realization differences, not statistically established scheme effects.

At fine resolution with WENO9, adding Smagorinsky changes fourth-hour LWP by -2.25 g/m² and peak w^2 by -0.027 m²/s². Third-moment marker RMSE changes from 0.0159 to 0.0313 m³/s³.

Coarse, Centered2 with Smagorinsky: sampled global q_t extrema over 0–4 h were -4.3385 to 15.0839 g/kg; 227 of 241 samples contained negative moisture. These are 60-second samples, not bounds on every timestep.

At coarse resolution with Smagorinsky, Centered2 minus WENO9 changes fourth-hour LWP by 5.52 g/m² and peak w^2 by 0.053 m²/s². Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

At coarse resolution with Smagorinsky, Centered2 minus WENO5 changes fourth-hour LWP by 6.59 g/m² and peak w^2 by 0.095 m²/s². Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Canonical, Centered2 with Smagorinsky: sampled global q_t extrema over 0–4 h were -7.6972 to 17.4041 g/kg; 231 of 241 samples contained negative moisture. These are 60-second samples, not bounds on every timestep.

At canonical resolution with Smagorinsky, Centered2 minus WENO9 changes fourth-hour LWP by -16.49 g/m² and peak w^2 by -0.027 m²/s². Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Breeze DYCOMS-II RF01: resolution and numerics

At canonical resolution with Smagorinsky, Centered2 minus WENO5 changes fourth-hour LWP by -15.33 g/m^2 and peak w^2 by $-0.006 \text{ m}^2/\text{s}^2$. Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Fine, Centered2 with Smagorinsky: sampled global q_t extrema over 0–4 h were -7.4007 to 18.0746 g/kg ; 234 of 241 samples contained negative moisture. These are 60-second samples, not bounds on every timestep.

At fine resolution with Smagorinsky, Centered2 minus WENO9 changes fourth-hour LWP by -9.06 g/m^2 and peak w^2 by $-0.072 \text{ m}^2/\text{s}^2$. Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

At fine resolution with Smagorinsky, Centered2 minus WENO5 changes fourth-hour LWP by -7.54 g/m^2 and peak w^2 by $-0.082 \text{ m}^2/\text{s}^2$. Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Fourth-hour metrics

Coarse / WENO9 / no SGS: LWP 49.87 g/m²; cloud 99.34%; peak w^2 0.395 m²/s²; Δq_t 0.019 g/kg; E 4.75 mm/s.

Coarse / WENO5 / no SGS: LWP 46.45 g/m²; cloud 98.91%; peak w^2 0.296 m²/s²; Δq_t 0.015 g/kg; E 4.70 mm/s.

Coarse / WENO9 / Smagorinsky: LWP 43.33 g/m²; cloud 99.07%; peak w^2 0.356 m²/s²; Δq_t 0.080 g/kg; E 5.21 mm/s.

Coarse / WENO5 / Smagorinsky: LWP 42.27 g/m²; cloud 98.52%; peak w^2 0.314 m²/s²; Δq_t 0.047 g/kg; E 4.93 mm/s.

Coarse / Centered2 / Smagorinsky: LWP 48.85 g/m²; cloud 99.44%; peak w^2 0.409 m²/s²; Δq_t 0.214 g/kg; E 7.27 mm/s.

Canonical / WENO9 / no SGS: LWP 61.69 g/m²; cloud 99.91%; peak w^2 0.521 m²/s²; Δq_t 0.003 g/kg; E 4.32 mm/s.

Canonical / WENO5 / no SGS: LWP 59.61 g/m²; cloud 99.92%; peak w^2 0.467 m²/s²; Δq_t 0.020 g/kg; E 5.19 mm/s.

Canonical / WENO9 / Smagorinsky: LWP 52.19 g/m²; cloud 99.37%; peak w^2 0.454 m²/s²; Δq_t 0.023 g/kg; E 5.24 mm/s.

Canonical / WENO5 / Smagorinsky: LWP 51.04 g/m²; cloud 98.95%; peak w^2 0.433 m²/s²; Δq_t 0.060 g/kg; E 5.24 mm/s.

Canonical / Centered2 / Smagorinsky: LWP 35.70 g/m²; cloud 99.59%; peak w^2 0.427 m²/s²; Δq_t 0.250 g/kg; E 6.57 mm/s.

Fine / WENO9 / no SGS: LWP 55.48 g/m²; cloud 99.95%; peak w^2 0.472 m²/s²; Δq_t 0.031 g/kg; E 5.34 mm/s.

Fine / WENO5 / no SGS: LWP 57.54 g/m²; cloud 99.90%; peak w^2 0.483 m²/s²; Δq_t 0.007 g/kg; E 5.24 mm/s.

Fine / WENO9 / Smagorinsky: LWP 53.23 g/m²; cloud 99.89%; peak w^2 0.445 m²/s²; Δq_t 0.036 g/kg; E 5.09 mm/s.

Fine / WENO5 / Smagorinsky: LWP 51.71 g/m²; cloud 99.77%; peak w^2 0.455 m²/s²; Δq_t 0.034 g/kg; E 5.48 mm/s.

Fine / Centered2 / Smagorinsky: LWP 44.17 g/m²; cloud 99.98%; peak w^2 0.373 m²/s²; Δq_t 0.154 g/kg; E 6.44 mm/s.

Fourth-hour metrics

Peak w^2 is the maximum of the fourth-hour mean native-face profile; entrainment includes subsidence. These are single-seed results.

Fine-grid averaging-window comparison

Compare the same statistic across averaging windows. Dimensional third moment and skewness differ in normalization as well as the windows used in the two papers. All minima below are over 600–760 m and may occur at different heights. Skewness is the ratio of time-averaged moments.

fine_none_weno9, 2-4 h: minimum w'^3 -0.0497 m³/s³; minimum skewness -0.1503.

fine_none_weno9, 3-4 h: minimum w'^3 -0.0622 m³/s³; minimum skewness -0.1928.

fine_none_weno5, 2-4 h: minimum w'^3 -0.0433 m³/s³; minimum skewness -0.1334.

fine_none_weno5, 3-4 h: minimum w'^3 -0.0372 m³/s³; minimum skewness -0.1109.

fine_smagorinsky_weno9, 2-4 h: minimum w'^3 -0.0419 m³/s³; minimum skewness -0.1394.

fine_smagorinsky_weno9, 3-4 h: minimum w'^3 -0.0304 m³/s³; minimum skewness -0.1026.

fine_smagorinsky_weno5, 2-4 h: minimum w'^3 -0.0163 m³/s³; minimum skewness -0.0538.

fine_smagorinsky_weno5, 3-4 h: minimum w'^3 -0.0160 m³/s³; minimum skewness -0.0521.

fine_smagorinsky_centered2, 2-4 h: minimum w'^3 -0.0236 m³/s³; minimum skewness -0.1206.

fine_smagorinsky_centered2, 3-4 h: minimum w'^3 -0.0272 m³/s³; minimum skewness -0.1309.

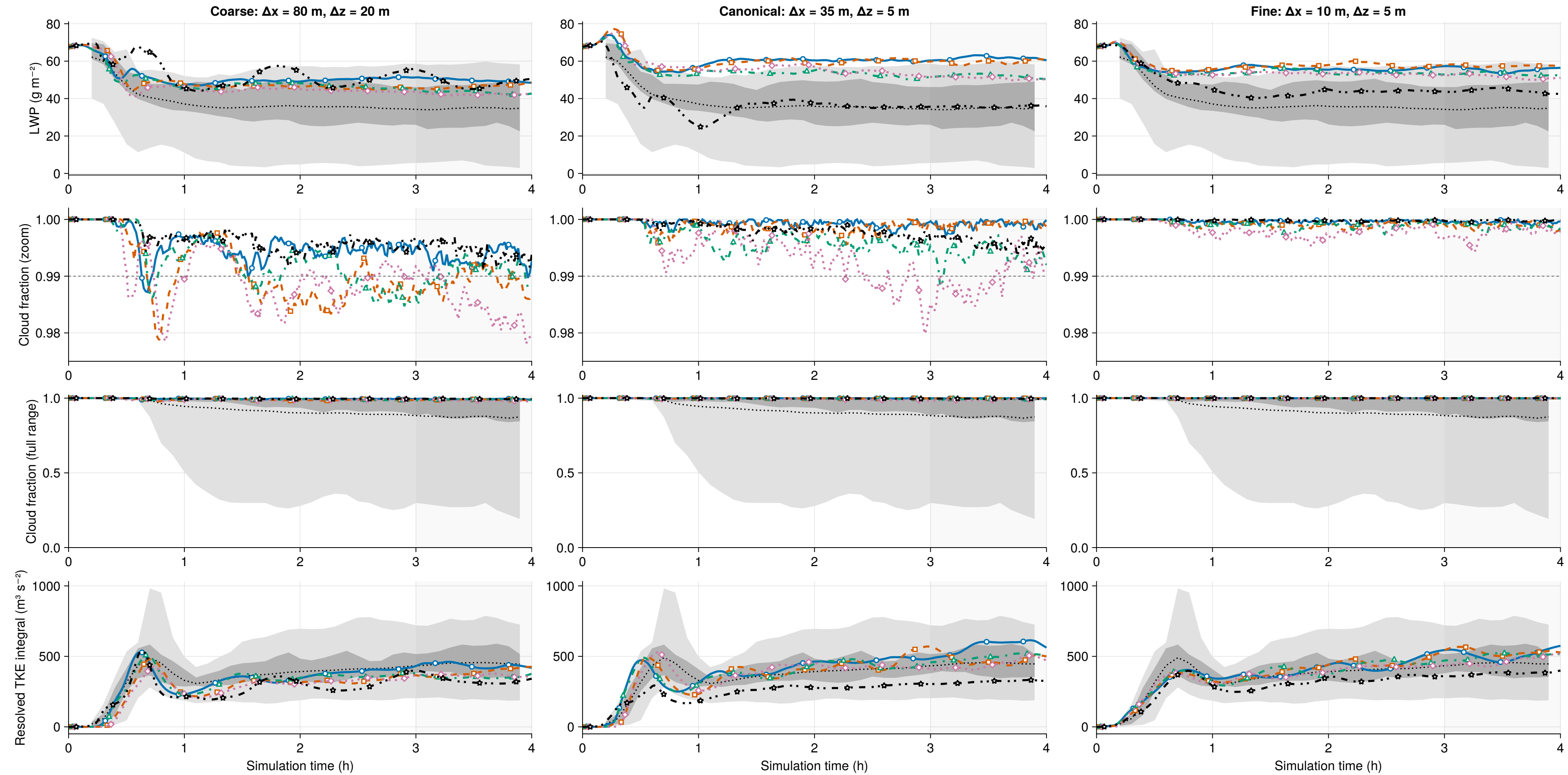
How to read these comparisons

All scientific figures are generated in Julia with CairoMakie. Numerical configurations use fixed Okabe-Ito colors, distinct dashes and marker shapes; different resolutions occupy panels with shared scales. The cloud-fraction zoom follows the simulated range, while a separate full-range panel preserves the paper ensemble spread.

Light gray is the paper's model-ensemble minimum–maximum; darker gray is its interquartile range. These are not observational uncertainty bands or validation tolerances. The mean and envelope were digitized from Figure 2 at approximately 0.2–3.9 h. Original pixel coordinates, calibration, extraction settings and an overlay are retained. Subpixel early cloud-fraction bands are left missing. Only rendered cloud-fraction band edges are clipped to the physical 0–1 range; CSV values preserve raw digitization. The zoom panel omits the paper bands to avoid magnifying raster-edge uncertainty.

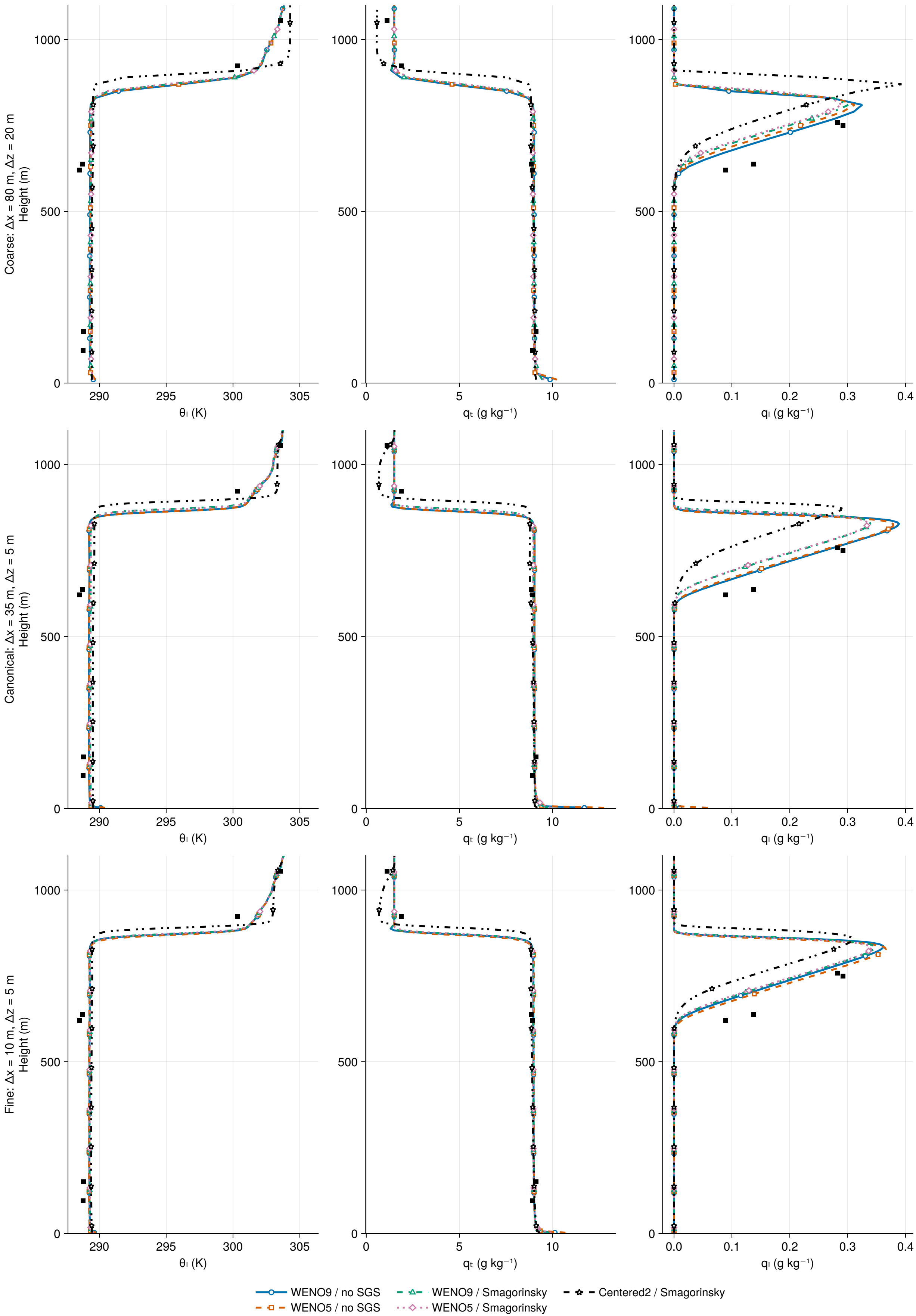
Paper observations are black markers in profile plots. Measurement error bars were not digitized. Numerical configurations use redundant color, line and symbol cues; resolution panels share scales.

Cloud persistence and turbulence evolution



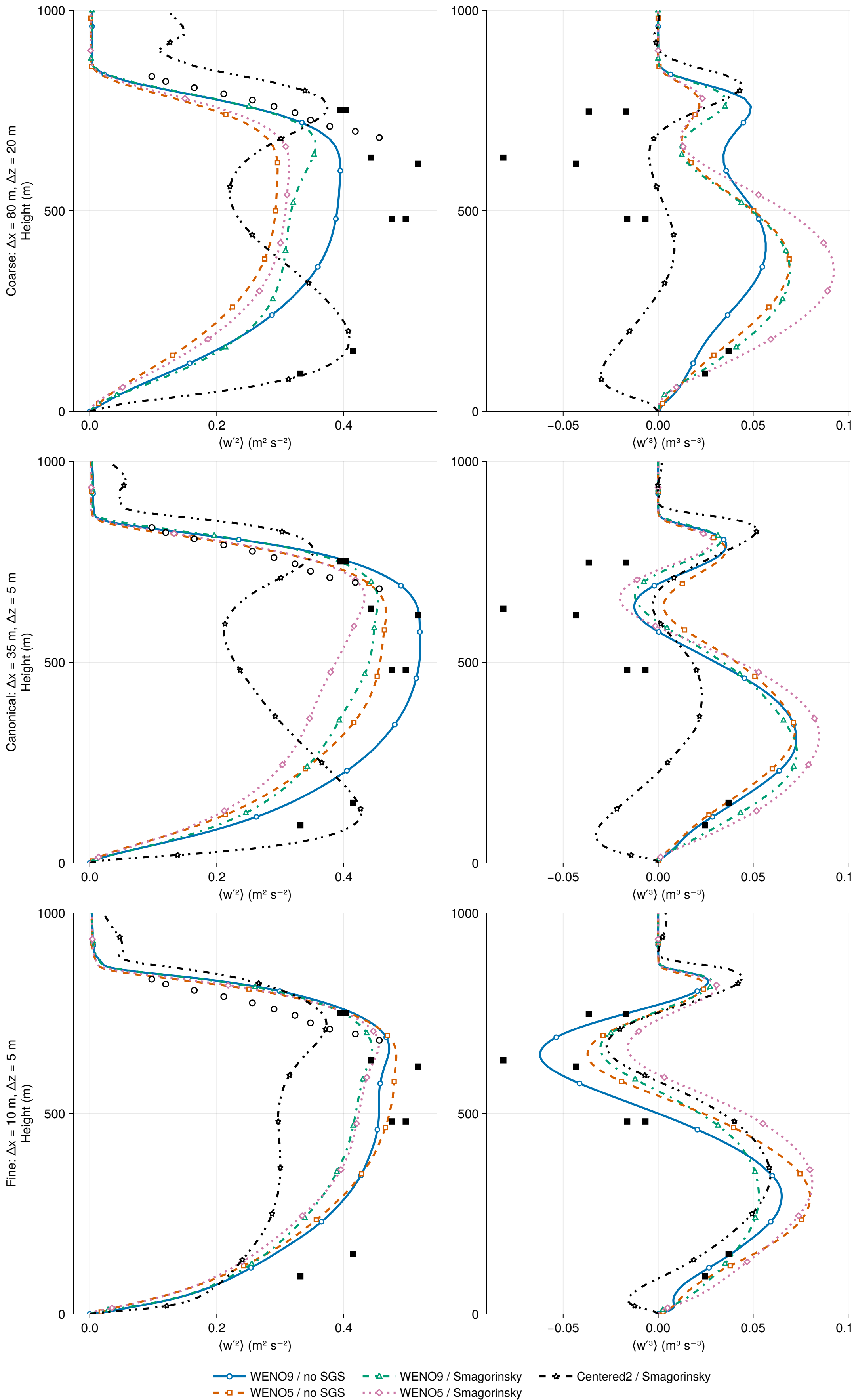
Gray bands: digitized MODEL spread (Stevens et al. 2005, Fig. 2), not observational uncertainty. Cloud-fraction zoom shows Breeze curves; the full-range row shows the paper envelope. Unresolved early shading is omitted. TKE is resolved only. All plots: Julia/CairoMakie.

Fourth-hour thermodynamic structure



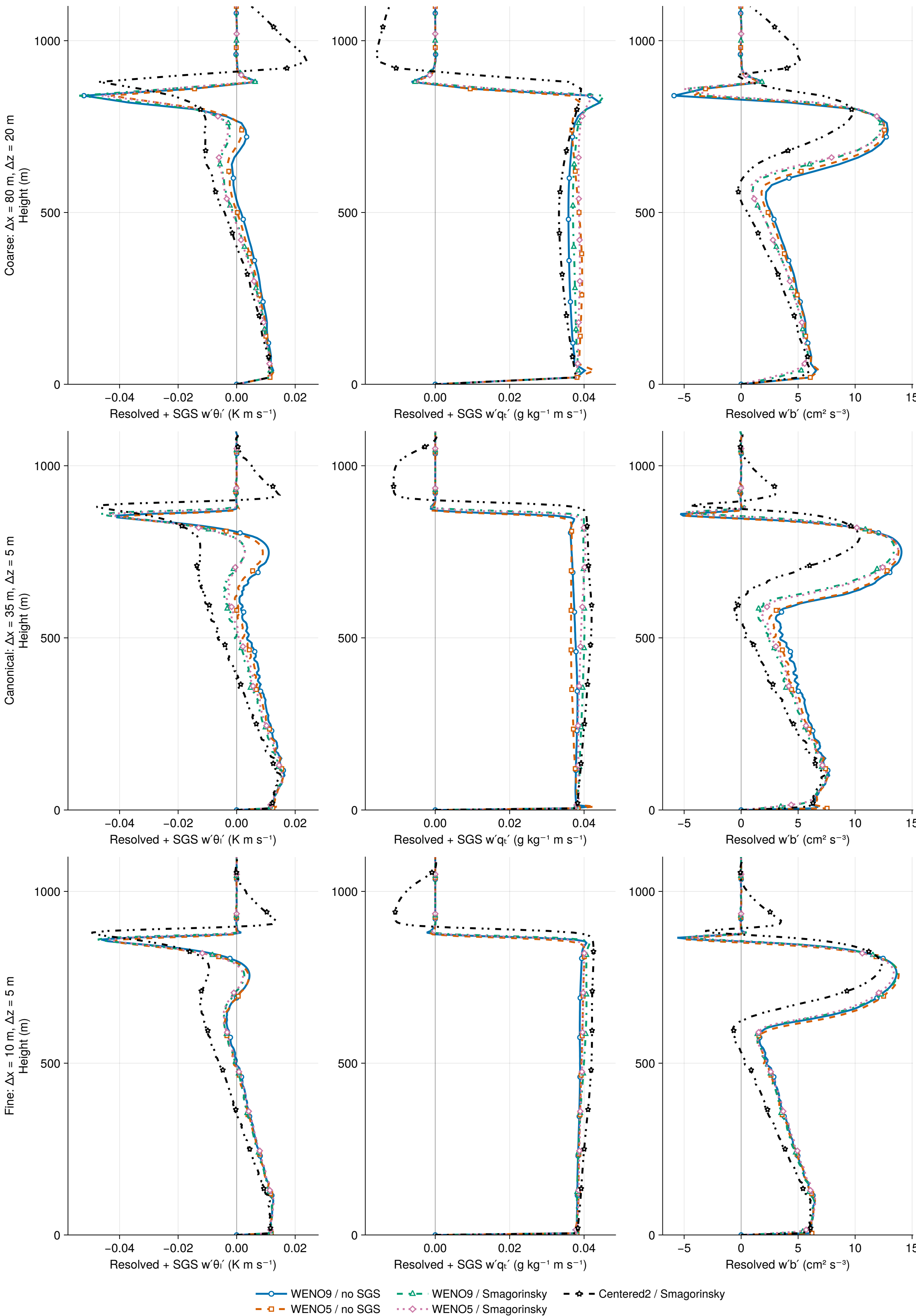
Fourth-hour means combine the two half-hour profile averages ending at 12600 and 14400 s. Black squares: retained paper Fig. 4 observations; measurement error bars were not digitized. Shared axis scales across resolutions.

Fourth-hour vertical-velocity moments



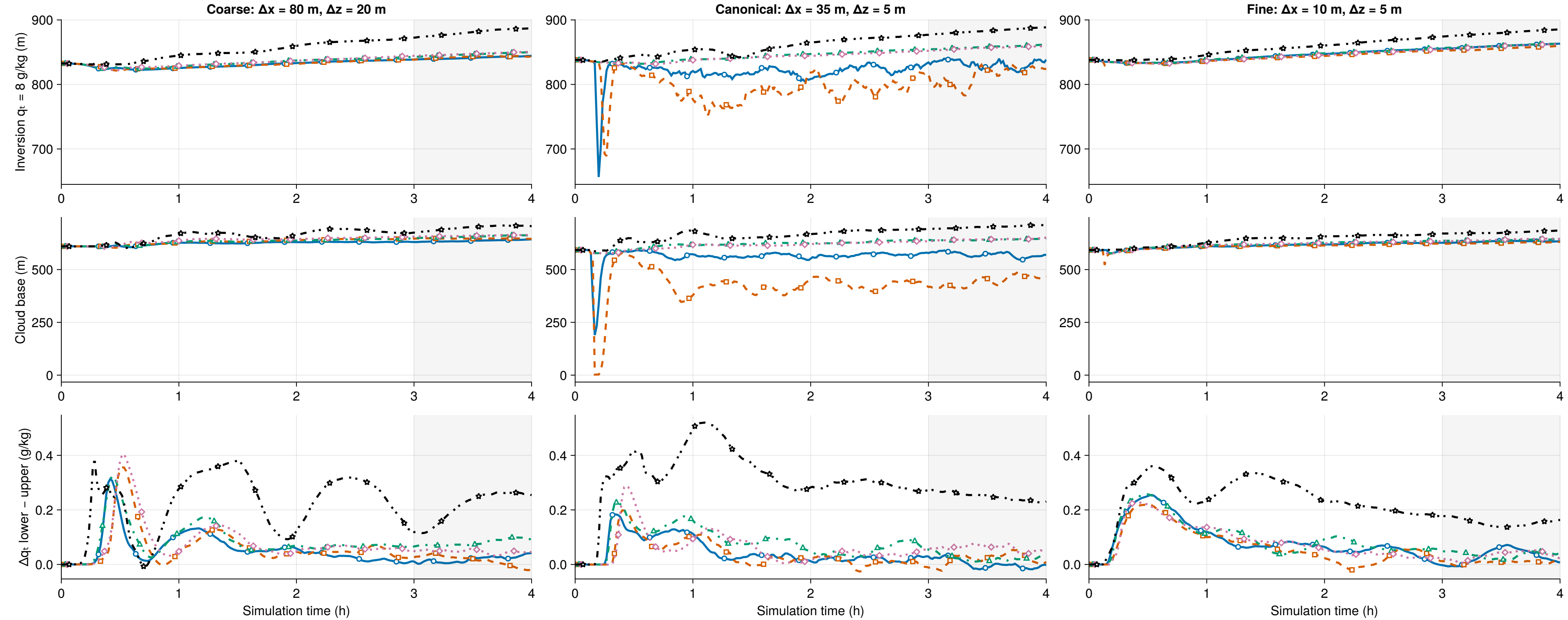
Fourth-hour means on native w faces. Black squares: retained in-situ observations; black open circles: radar (paper Fig. 5). Error bars were not digitized. Colors, line patterns and marker shapes identify numerical configurations; rows identify resolutions.

Fourth-hour scalar transport and buoyancy production



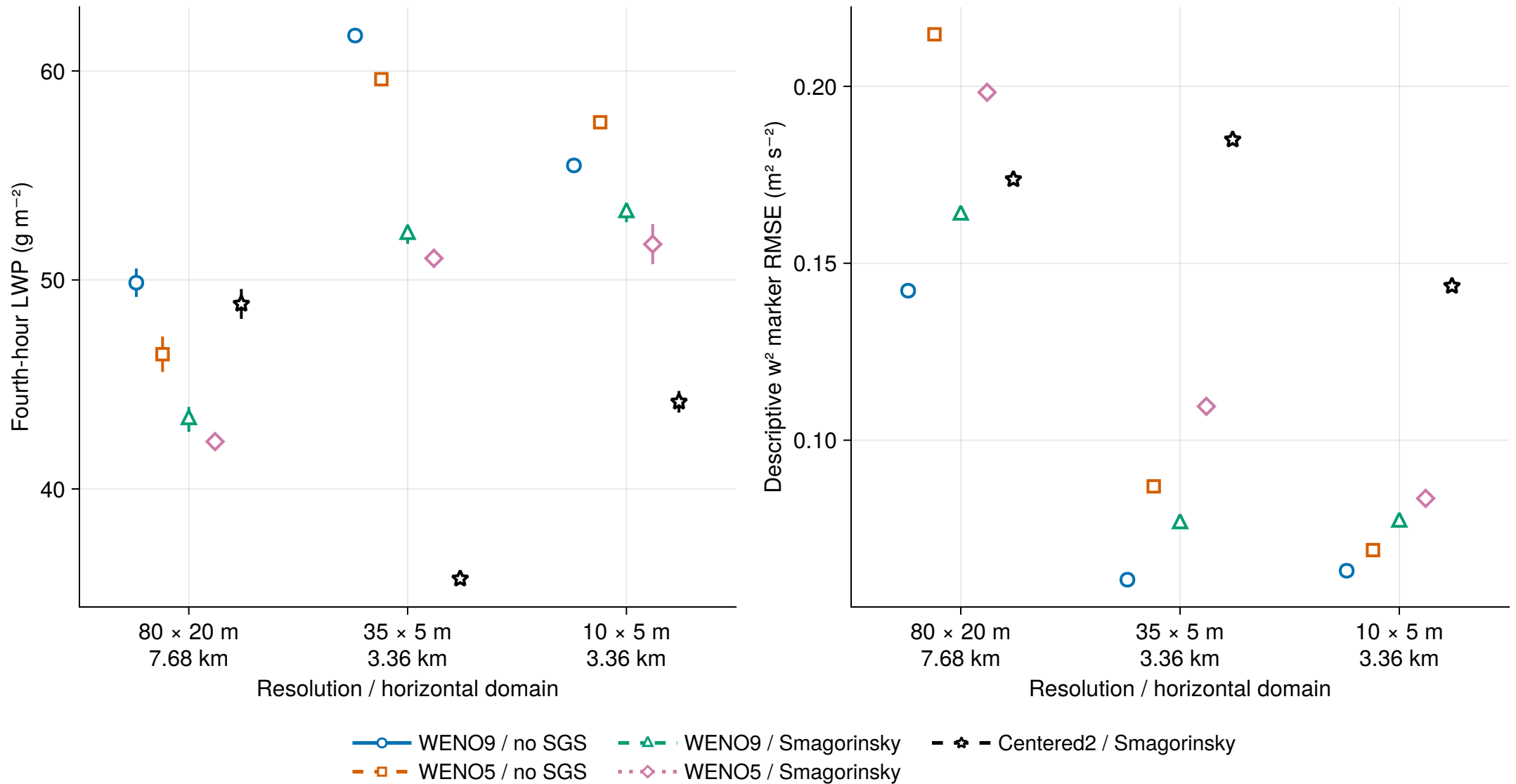
Scalar panels add resolved covariance and explicit SGS flux; buoyancy production is resolved only. Numerical transport is not separately measured, so resolved + SGS is not a complete discrete flux budget.

Cloud boundaries and decoupling



Cloud-base height is conditional on cloudy columns. Inversion height uses the interpolated q_t contour. $\Delta q_t = \text{mean}(100\text{--}200 \text{ m}) - \text{mean}(700\text{--}800 \text{ m})$, using exact geometric layer overlap.

Resolution and numerical choices



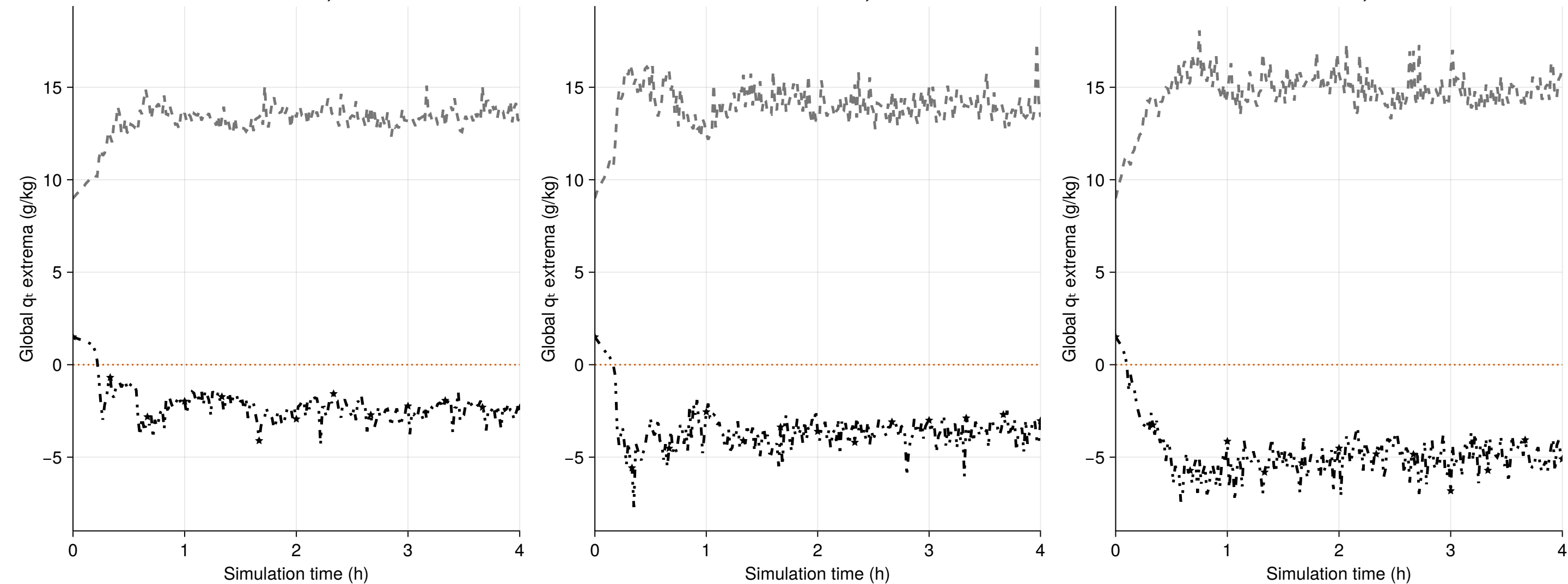
Only completed cases appear. LWP bars span two half-hour means, not confidence intervals. Marker RMSE is an unweighted descriptive distance to retained in-situ observations, not an uncertainty-normalized validation score.

Unbounded Centered2 + Smagorinsky: moisture extrema

Coarse: $\Delta x = 80$ m, $\Delta z = 20$ m

Canonical: $\Delta x = 35$ m, $\Delta z = 5$ m

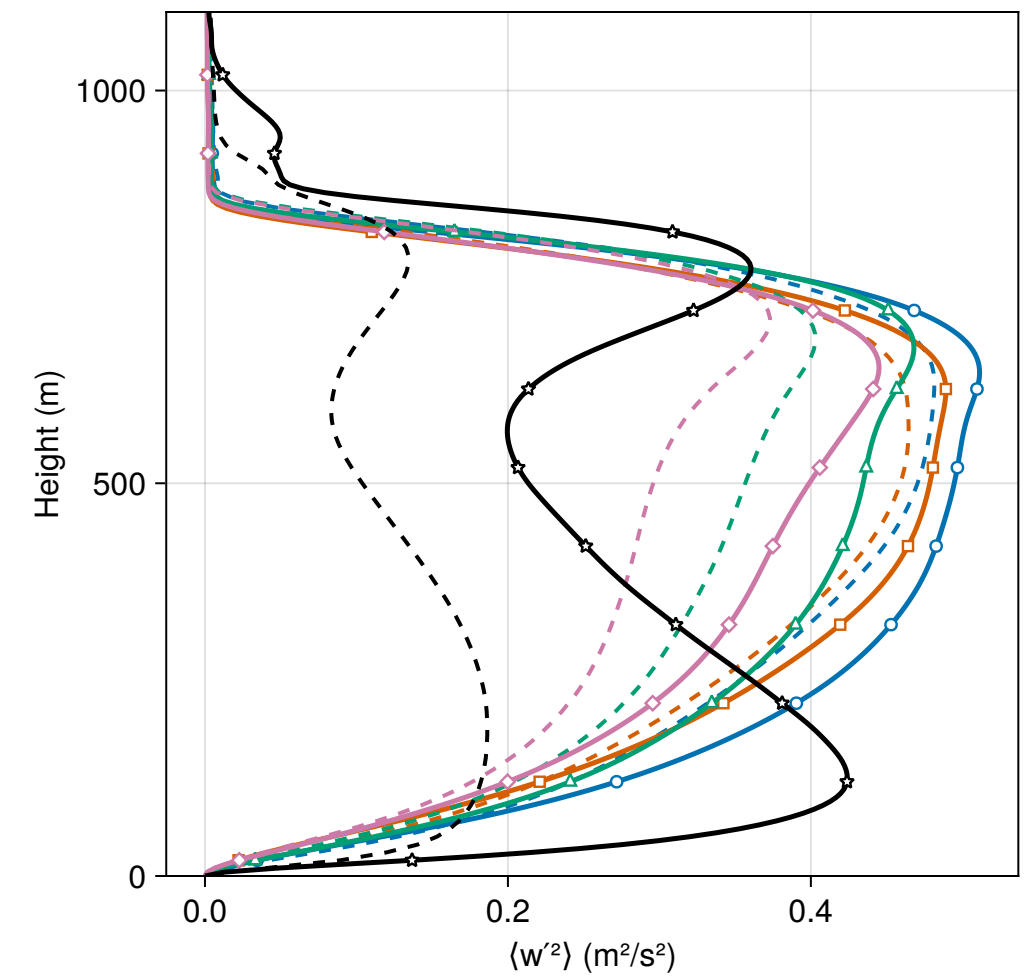
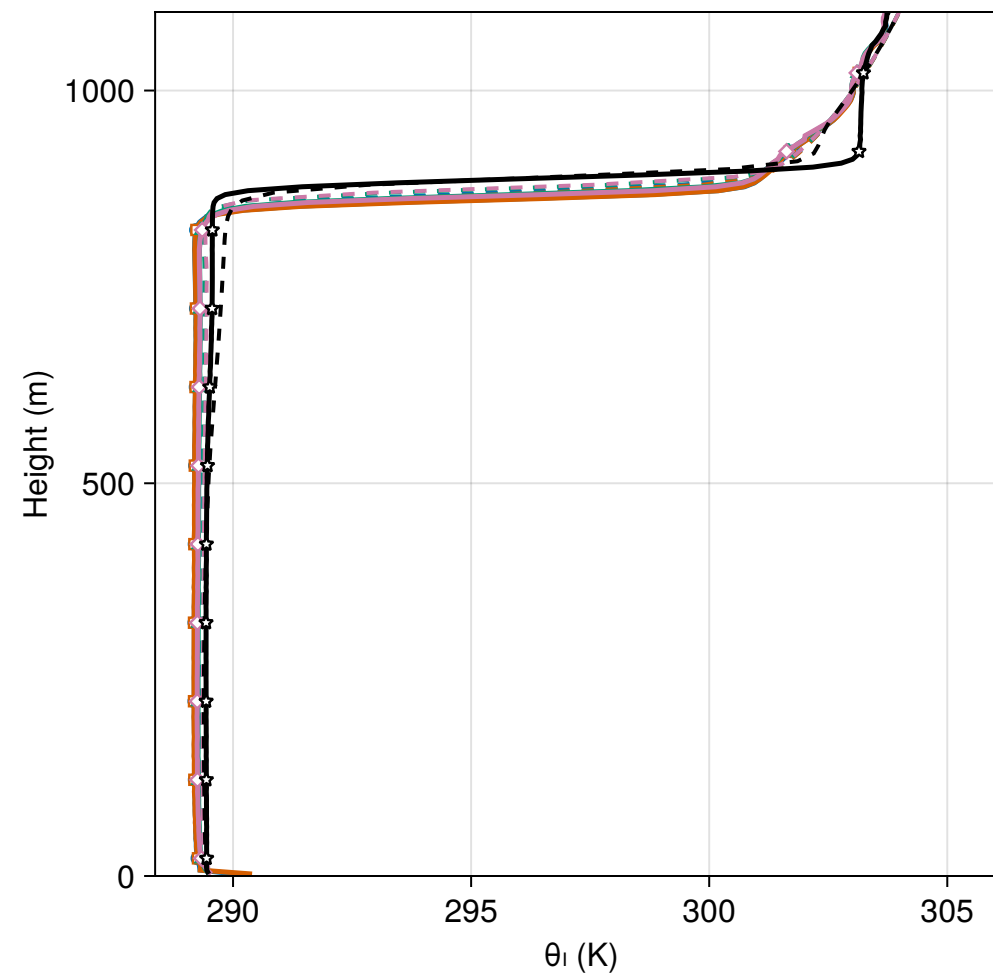
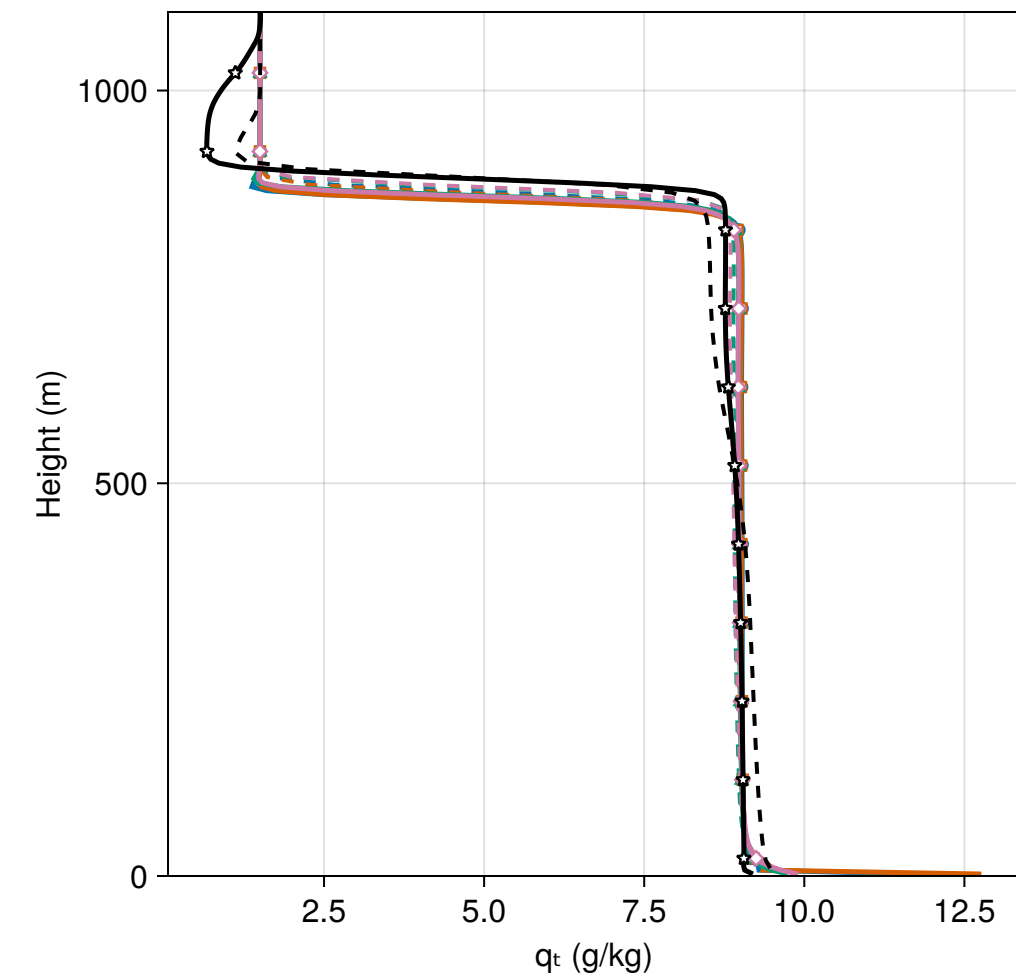
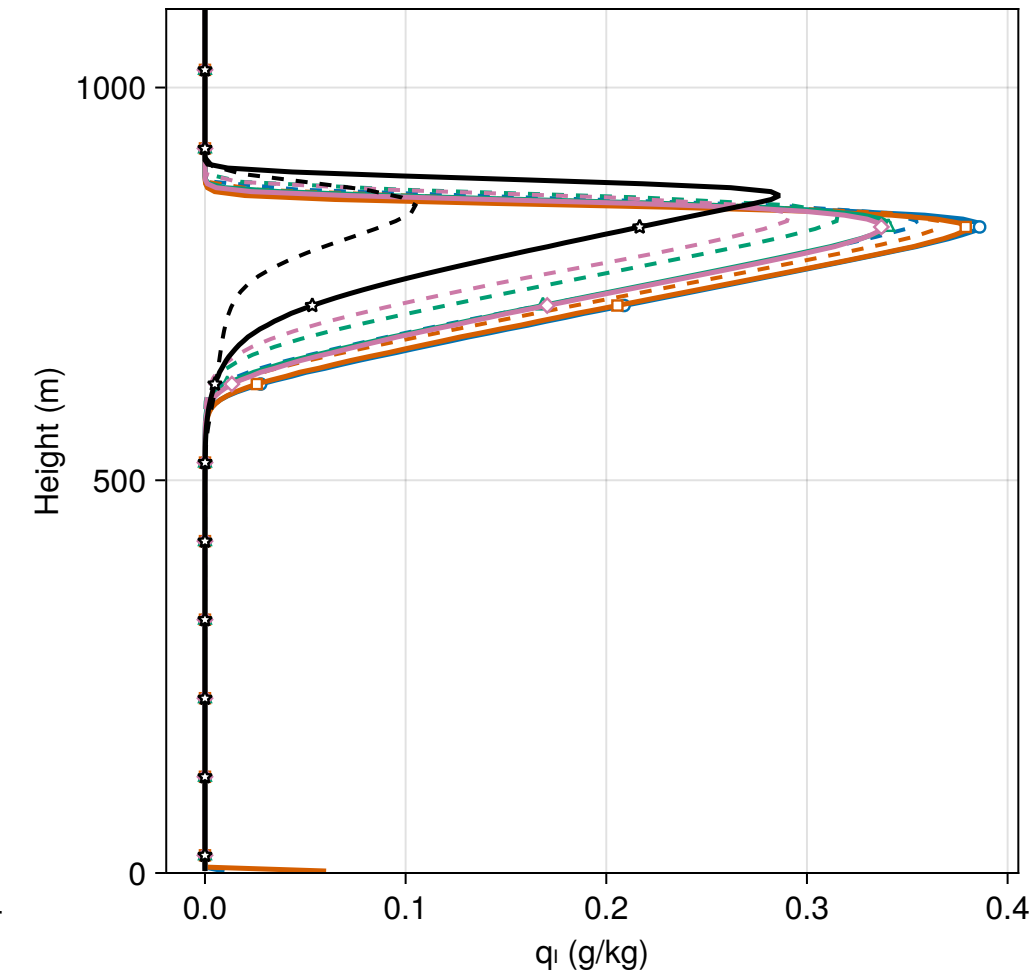
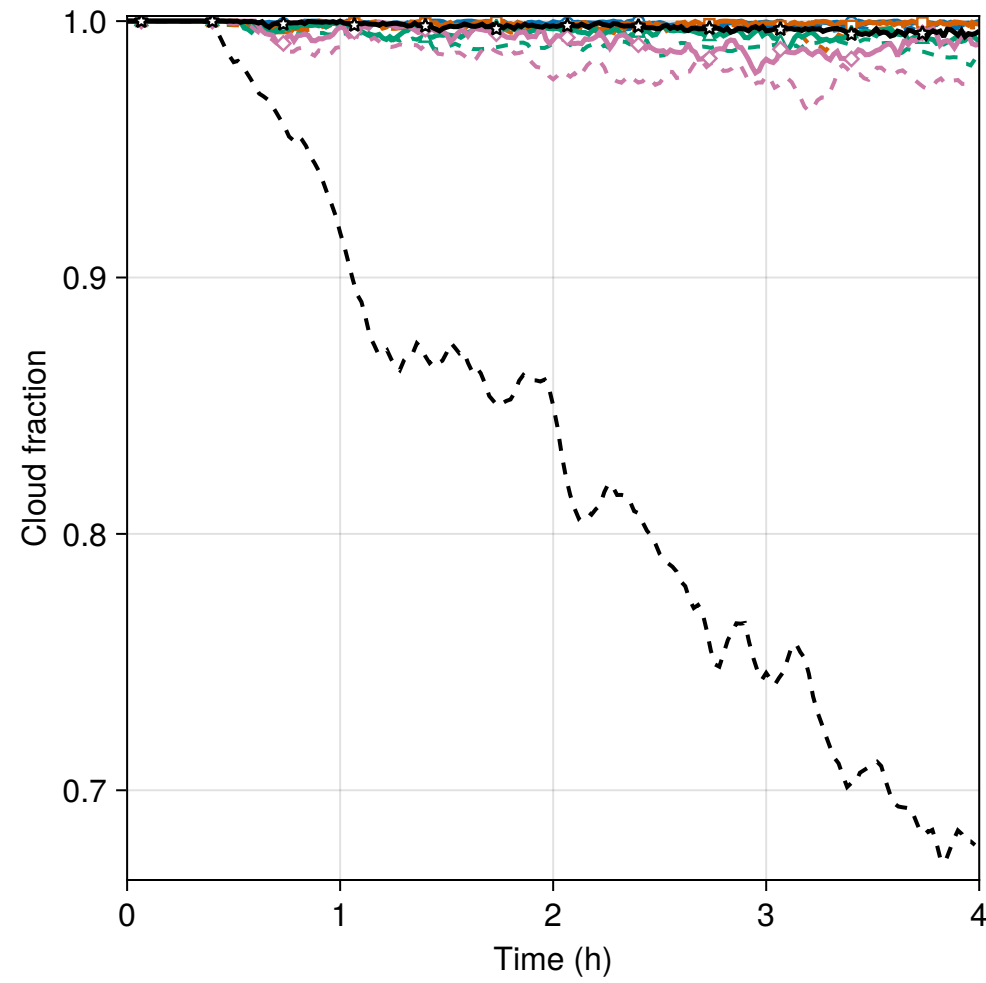
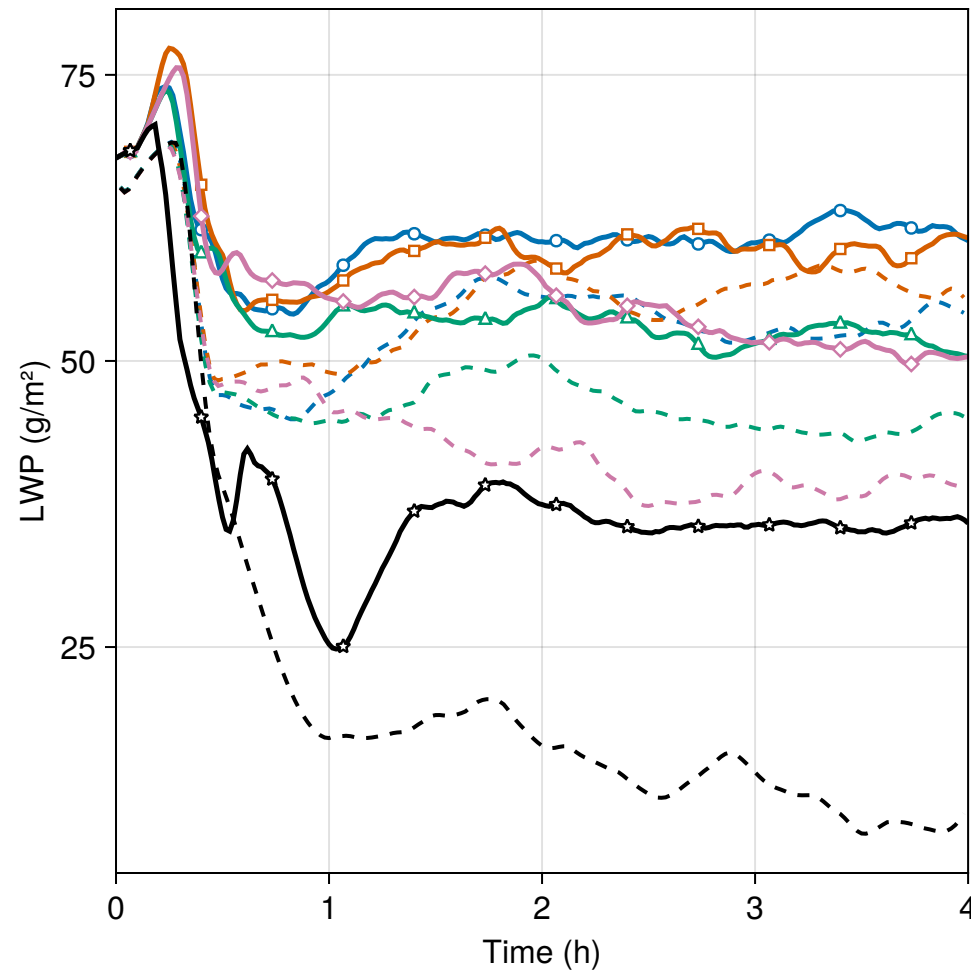
Fine: $\Delta x = 10$ m, $\Delta z = 5$ m



- · · · Minimum q_t (star markers) - - - Maximum q_t · · · · Zero total water

Global spatial extrema sampled every 60 s; excursions between samples are not bounded by these curves.
Negative total water is physically inadmissible. Finite output does not establish physical validity; no clipping or WENO fallback was applied.

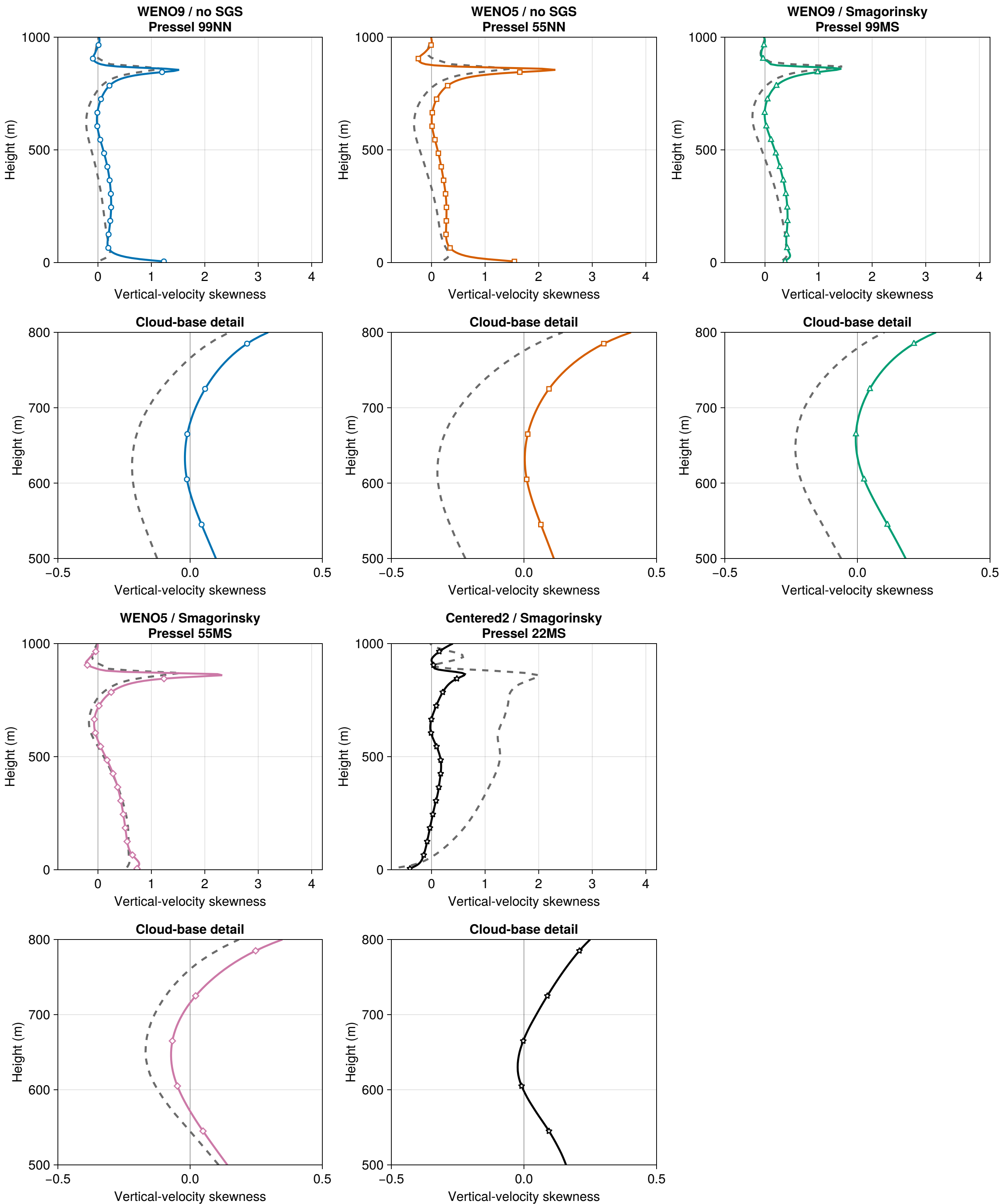
Canonical grid: Breeze and Pressel et al. (2017)



-- WENO9 / no SGS (99NN) -- WENO9 / Smagorinsky (99MS) -- Centered2 / Smagorinsky (22MS) --- Pressel model
-- WENO5 / no SGS (55NN) -- WENO5 / Smagorinsky (55MS) — Breeze

35 m horizontal / 5 m vertical; same domain. Profiles average 2–4 h. Cloud-fraction axis zoomed. Only completed Breeze cases shown.
 Scheme-family comparison: Cs, Pr, CFL, thermodynamics, surface treatment and moisture handling differ; this is not a replication.

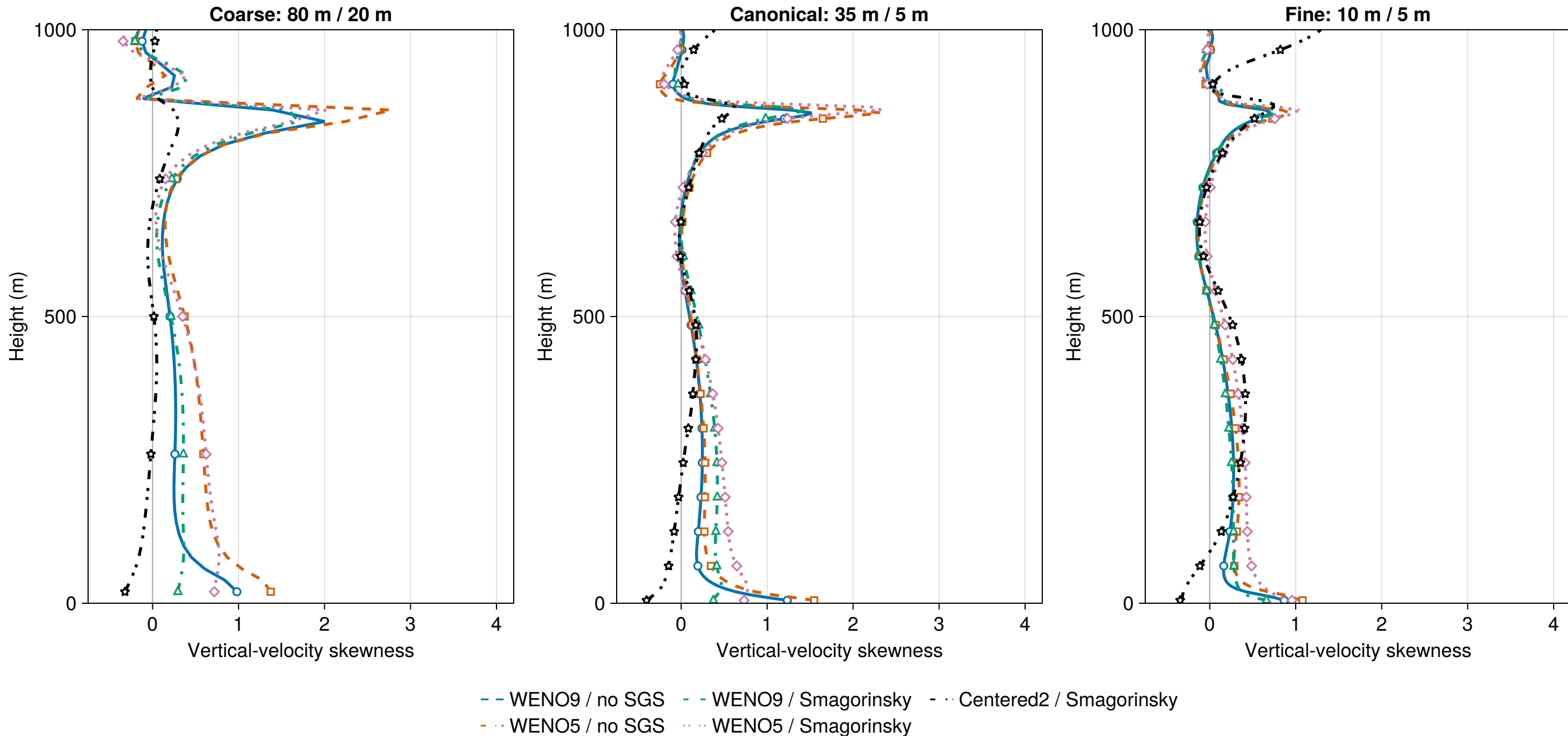
Breeze and Pressel Figure 7: skewness, hours 2–4



— Breeze: colored solid + symbols - - Pressel model: gray dashed

Canonical grid: $\Delta x = 35$ m, $\Delta z = 5$ m. Breeze uses the ratio of time-averaged central moments.
 Pressel normalization/averaging order is unverified; these are model curves, not observations.
 Top: 0–1000 m. Bottom: magnified 500–800 m. Low-variance levels masked. Model settings differ.

Breeze skewness across resolutions: hours 2–4



Ratio of 2–4 h averaged central moments; not the time average of instantaneous skewness.
 Only completed exports shown. Coarse horizontal domain is larger. Variance $\leq 10^{-6} \text{ m}^2/\text{s}^2$ is masked; full-height values are in CSV.

Methods and limitations

Experimental design

Three grids: coarse $96 \times 96 \times 75$ at $80 \times 80 \times 20$ m in a $7.68 \times 7.68 \times 1.5$ km domain; canonical $96 \times 96 \times 300$ at $35 \times 35 \times 5$ m in $3.36 \times 3.36 \times 1.5$ km; fine $336 \times 336 \times 300$ at $10 \times 10 \times 5$ m in the same canonical domain. Fine $dz=5$ m is the stated implementation assumption. Each grid crosses WENO9/WENO5 with no closure/Smagorinsky-Lilly ($C_s=0.16$, $Pr=1$). WENO moisture uses bounds=(0,1) at both orders; order changes apply to all advected variables. Three additional runs apply Centered(order=2) to momentum and all scalars, including moisture, with the same Smagorinsky-Lilly settings: 15 cases total. Centered moisture is unbounded, so comparisons with WENO change scheme family and bounding as well as order. Finite diagnostics and moisture extrema must be checked before interpreting these cases.

Physics and initial conditions

Breeze uses anelastic dynamics, warm-phase saturation adjustment, fixed surface sensible/latent heat fluxes $15/115$ W/m², drag coefficient 0.0011, divergence $D=3.75 \times 10^{-6}$ s⁻¹, geostrophic winds (7,-5.5) m/s and the RF01 idealized longwave radiation. Initial inversion is 840 m, $\theta_l=289$ K and $q_t=9$ g/kg below, $\theta_l=297.5+(z-840)^{(1/3)}$ K and $q_t=1.5$ g/kg above. No drizzle or droplet sedimentation. All runs use Float32, seed 123 and four simulated hours. The fixed-flux moisture conversion uses $L_v=2.47 \times 10^6$ J/kg; Breeze retains its default thermodynamic latent-heat law with dry-air $c_p=1015$ J/(kg K), which differs from a strictly constant-mixture- c_p implementation.

Averaging and grid locations

Profiles are averages of instantaneous horizontally reduced statistics. The fourth hour combines the equal-duration bins ending at 12,600 and 14,400 s; it excludes the bin ending at 10,800 s. Scalar means are on centers; w variance and third central moment remain on native vertical faces. The third moment has units m³/s³, not normalized skewness. Series means use trapezoidal integration of 60 s samples between 10,800 and 14,400 s. Two half-hour LWP means give a descriptive temporal spread, not a confidence interval.

Methods and limitations

Cloud and entrainment definitions

Column cloud fraction uses $q_l > 10^{-6}$ kg/kg in any cell. Cloud-base means exclude clear columns. Inversion height uses the first downward $q_t = 8$ g/kg crossing with linear interpolation; its valid fraction is checked. The $\theta_{\text{ref}} = 295$ K contour and the nearest-cell height used internally by radiation are retained separately. Fourth-hour entrainment is $[z_i(4h) - z_i(3h)]/3600 + D \cdot \text{mean}(z_i)$, including subsidence. Decoupling is mean q_t over 100–200 m minus mean q_t over 700–800 m, with exact layer-overlap weights.

Reference data and comparison limits

Stevens et al. (2005), DOI 10.1175/MWR2930.1, supplies the RF01 comparison. Figure 2 is a model ensemble mean, not observations or an acceptance band; the paper has no observed LWP time series. Observed cloud cover exceeds 99%. Figure 4 and 5 markers are partial digitizations, with source pixels retained and error bars not extracted. Marker RMSE is an unweighted descriptive distance to retained in-situ points; it is not a likelihood, uncertainty-normalized score, or formal validation criterion. Radar markers are plotted separately. Resolved TKE is plotted against the historical ensemble for context; a prognostic SGS-TKE contribution is unavailable for these closures.

What this design can establish

The coarse domain has 5.22 times the canonical horizontal area, so coarse-versus-canonical differences combine changes in horizontal spacing, vertical spacing and domain size. Fine-versus-canonical holds domain and vertical spacing fixed. Closure/order comparisons share a seed within a grid, but one seed gives no ensemble uncertainty and different grids do not share identical perturbation fields. UCLA-0 in the paper disables scalar SGS mixing while retaining momentum treatment; it is not identical to Breeze closure=nothing. Neither matching LWP nor agreement with one fine simulation establishes convergence.

Methods and limitations

Numerical and output audit

The original WENO runs use an immutable source snapshot with per-file hashes and a pinned Julia 1.12.6 workspace. The Centered2 extension uses a separate versioned runner and provenance while preserving the original source. GPU smoke runs checked all grids and the diagnostic output path before production. Exports require completed jobs, end-of-run sentinels, finite numeric records, exactly nine profile times and 241 series times, 46 profile variables and 32 series variables. Native face/center coordinates and fixed fluxes are checked again locally. Replicated budget arrays, where present, are horizontally averaged by the exporter and their original shape/spread recorded. The TKE residual is a budget-closure residual, not a measured numerical dissipation.

Reproducibility and current run state

All active processing and figures use Julia. CairoMakie version 0.15.14; Julia 1.12.7. The companion Markdown/HTML report links to all CSVs, source hashes, digitization provenance and exact rebuild commands. Original restart checkpoints and JLD2 files remain on pcluster.

coarse_none_weno9: COMPLETED

coarse_none_weno5: COMPLETED

coarse_smagorinsky_weno9: COMPLETED

coarse_smagorinsky_weno5: COMPLETED

canonical_none_weno9: COMPLETED

canonical_none_weno5: COMPLETED

canonical_smagorinsky_weno9: COMPLETED

canonical_smagorinsky_weno5: COMPLETED

fine_none_weno9: COMPLETED

fine_none_weno5: COMPLETED

fine_smagorinsky_weno9: COMPLETED

fine_smagorinsky_weno5: COMPLETED

coarse_smagorinsky_centered2: COMPLETED

canonical_smagorinsky_centered2: COMPLETED

fine_smagorinsky_centered2: COMPLETED

Reference: Stevens et al. (2005), Monthly Weather Review 133, 1443–1462. DOI 10.1175/MWR2930.1.